Universal Information Structure Analysis: Toward Carrier-Agnostic Discovery of Recoverable Structure in Digital Artifacts

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Extended Zenodo Research Draft. This version substantially expands the earlier conference draft while preserving the same implementation, benchmark data, seeds, figures, and reported numerical results. The purpose of this document is to establish a complete research record and a technically explicit basis for subsequent peer-reviewed submission.

Abstract. Digital information is normally analyzed inside a declared representation: text is treated as text, audio as audio, video as video, and steganalysis as the detection of a covert modification within an already identified carrier. This paper proposes *Universal Information Structure Analysis (UISA)* as a broader inverse problem: given an arbitrary digital artifact, determine whether it contains statistically significant, reproducible, potentially recoverable information-bearing structure, and discover a representation in which that structure becomes measurable or decodable without assuming the carrier, alphabet, synchronization rule, or modality in advance. The proposed framework connects information theory, minimum description length (MDL), atypicality, signal processing, representation learning, multimodal inference, steganalysis, and canonical representation. It separates structure detection from claims about intent and treats representation discovery itself as an explicit search variable. A reproducible prototype is implemented using candidate transformations, description-length-inspired atypicality scores, synthetic motion and audio carriers, controlled structure injection, and a small learned cross-modality classifier. Revalidation reproduces the original benchmark exactly: the candidate scan gives motion null/structured means of 2.044/2.058 and audio null/structured means of 0.797/1.135, while the learned random-forest baseline obtains 0.825 accuracy and 0.9163 ROC-AUC on a held-out synthetic test set. The strong overlap in the candidate-scan experiment is not treated as a failure of the research program but as an important negative result: natural structure can itself be compressible, atypical, and highly informative, so a generic anomaly score cannot by itself distinguish intentional encoding from ordinary organization. The paper therefore defines a research architecture in which detection, representation pursuit, channel discovery, canonical recovery, residual analysis, and reproducibility are separate stages. The current evidence establishes a reproducible proof of concept, not a universal decoder.

Keywords: information structure; representation discovery; minimum description length; atypicality; multimodal signals; universal steganalysis; inverse problems; canonical representation; information theory; signal processing

1. Introduction

The central premise of this work is simple but unusually broad: an observable artifact does not come with its representation system attached. A byte sequence can be interpreted as text, samples, pixels, instructions, records, compressed data, or a mixture of these. A sequence of measurements can be interpreted as a physical trajectory, a control signal, a symbolic stream, or a projection of a higher-dimensional process. The useful question is therefore not always “what does this modality contain?” but “which representation makes the information-bearing structure recoverable?”

This distinction becomes important when the carrier is not selected from a conventional media taxonomy. Consider a performance by a dancer. The information available to an observer includes joint configurations, hand and finger configurations, gaze, torso orientation, stage position, velocity, acceleration, timing relative to music, recurrence of poses, transitions between poses, and relationships among all of these variables. None of these variables is intrinsically the “channel.” A message could be represented by a sequence of mudras, by relative joint angles, by positions on the stage, by temporal offsets to the beat, by coordinated cross-modal transitions, or by a combination. A conventional video steganalyzer starts from a very different assumption: it asks whether a known or related embedding process has modified a video carrier. UISA asks whether the observable contains a recoverable information structure at all, and then asks which state space exposes it.

The same problem appears outside audiovisual media. A mechanical system has discrete and continuous states. A biological sequence has a finite alphabet but can be represented at many scales, from bases to codons to genes to regulatory motifs. A musical performance contains pitch, duration, timbre, rhythm, instrument identity, dynamics and relations between performers. A compiler transforms surface syntax into progressively more canonical representations because the same underlying computation can be expressed in many forms. In each case, the analyst is effectively solving an inverse representation problem.

The present paper does not claim that this problem has been solved. Instead, it formalizes a research direction and supplies an executable scaffold. The deliberately modest experimental design is important. If a universal detector is defined too loosely, a model can be declared successful simply because it detects anything unusual. But natural language, music, dance, engineered motion and ordinary video are already highly structured. A detector that labels every compressible or predictable artifact as “encoded” is not useful. The challenge is to identify structure that is both statistically supported and potentially recoverable, while controlling for natural structure, model selection, multiple testing, and representation complexity.

The work is motivated by several observations. First, information can reside in higher-order and cross-modal relationships rather than in marginal distributions. Second, the detector itself defines a search strategy; if it stops after finding one easy channel, it may never discover additional channels. Third, a representation can be deliberately chosen so that the encoded object appears as an ordinary structured artifact. Fourth, the finite state space of a digital artifact does not make the inverse search computationally trivial: representation hypotheses can be numerous, and each hypothesis can have a nontrivial evaluation cost. Finally, statistical evidence of structure is not evidence of human intention.

The resulting research question is therefore: **Can an automated system discover statistically defensible, representation-independent evidence of recoverable information structure in arbitrary digital artifacts, and progressively infer the representation or channel through which that structure can be decoded?**

The remainder of the paper develops this question from information-theoretic foundations through prior work, formalization, architecture, experimental design, negative and positive results, statistical issues, computational complexity, and a roadmap toward real multimodal data.

2. Background: Information Is Not Identical to Its Carrier

Classical information theory begins with a source, a channel, and a receiver. In engineering practice these abstractions are powerful precisely because the representation is normally agreed upon before decoding begins. Bits are bits because the receiver knows how bits are grouped and interpreted. Audio samples are samples because the receiver knows the sampling process. Packets are packets because a protocol defines their boundaries and semantics.

UISA considers the complementary case in which the receiver does not know the representation. Let X denote the observed artifact. There may exist many transformations R_1, R_2, …, R_K under which different structures become visible. The problem is not merely to estimate parameters of a known model; it is to search over the model coordinate system itself. This makes UISA an inverse problem with an open representation space.

A useful analogy is coordinate selection in physics or signal processing. A waveform may look complicated in the time domain and sparse in the frequency domain. A periodic process may be obvious under autocorrelation. A dynamical system may become simple in a state-space representation. A grammar may be difficult to see in raw character statistics but obvious in a parse tree. The transformation does not create the underlying regularity; it exposes it.

This distinction also explains why entropy alone is insufficient. High entropy can be consistent with information-rich structured data, while low entropy can arise from a trivial repetitive process. Shannon entropy is a property of a distribution under a chosen observation model; it is not a universal semantic detector. Likewise, a z-score or anomaly score is meaningful only relative to a null model and a search procedure.

The relevant hierarchy is therefore broader than first-order statistics. One can inspect individual values, distributions, correlations, higher-order dependencies, temporal transitions, cross-modal transitions, latent state trajectories, and transformations that produce compact or predictive representations. UISA treats these as candidate levels in a representation search rather than as mutually exclusive definitions of information.

This viewpoint is particularly relevant to the user-generated encoding systems that motivated the present program. In those systems, arbitrary information may be projected into text, music, image, video, physical motion, or combinations thereof. The encoding alphabet may itself consist of phrases, musical structures, biological sequences, visual states, or other complex symbols. Once arbitrary state systems are admitted as alphabets, the inverse problem becomes one of discovering which state variables and transition rules constitute the effective symbol stream.

The scientific difficulty is that the hypothesis space becomes open-ended. A successful system must therefore avoid the impossible goal of literally trying every representation. Instead, it needs a principled search over a structured family of candidate representations, with complexity penalties, held-out evidence, and stopping criteria that discourage arbitrary overfitting.

3. Related Work and Literature Review

3.1 Shannon information theory. Shannon's 1948 framework established entropy, mutual information, channel capacity, and the separation between source and channel coding. UISA inherits the distinction between information-bearing dependencies and transmission mechanisms but changes the receiver's problem: the receiver may not know which representation corresponds to the source alphabet or channel coordinates. Mutual information remains useful as a recovery metric, but it cannot by itself specify the correct representation.

3.2 Minimum Description Length. Rissanen's MDL principle provides a natural language for representation discovery. A model is valuable when the combined description of the model and data is shorter than competing explanations. In UISA, this suggests scoring candidate representations by the reduction in description length relative to an appropriate null. The important qualification is that the representation itself has a cost. Without that penalty, a sufficiently flexible representation can encode the observation nearly verbatim and appear maximally successful.

3.3 Algorithmic information and compression. Kolmogorov-style reasoning motivates the broader idea that regularities correspond to shorter descriptions, although exact Kolmogorov complexity is not computable. Practical systems therefore rely on computable compressors, predictive models, learned representations, or restricted model classes. UISA adopts the same operational stance: the score is a model-relative approximation, not a claim to compute algorithmic complexity.

3.4 Atypicality and data discovery. Sabeti and Høst-Madsen developed atypicality-based methods for discovering interesting real-valued data using coding-theoretic comparisons rather than simple outlier distance. This is closely related to UISA because it treats "interestingness" as a relation between data and a model of typical data. The extension proposed here is to make the representation transform itself part of the discovery process and to incorporate multiple modalities and candidate channels.

3.5 Universal steganalysis. Universal steganalysis is the closest established neighboring field. Arivazhagan et al. demonstrated a universal image steganalyzer spanning content-independent and content-adaptive algorithms. Cao, Tian and Li extended the universal-steganalysis idea to several HEVC intra-frame embedding domains using IPM maps and a convolutional residual network. These are important precedents for the term "universal" within a declared carrier domain. UISA is broader in the inverse-problem sense: it does not assume that the artifact is a stego object, that the carrier is video, or that a known embedding family exists.

3.6 Multimodal steganalysis. Recent audiovisual work is particularly relevant to UISA because it shows that multimodal detection can be deceptive if a model learns the wrong modality. Paudel, Jha and Parakh report that unimodal detectors for split-payload audiovisual steganography can approach random performance, while apparent multimodal improvement can be driven largely by the video stream. This supports an explicit requirement for UISA: cross-modal evidence should be tested for incremental explanatory value rather than assumed to be jointly informative.

3.7 Representation learning and self-supervised learning. Modern representation learning provides methods for discovering latent coordinates that make prediction or discrimination easier. UISA is related but has a different evaluation target. A learned embedding is not automatically a useful information representation for arbitrary artifacts; the representation must be tested for reproducibility, recoverability, complexity and residual structure. The desired system is closer to an adaptive scientific instrument than to a fixed classifier.

3.8 Signal processing and blind source separation. Classical signal processing already contains a rich literature on blind source separation, spectral estimation, time-frequency analysis, synchronization and system identification. UISA should be viewed as an umbrella problem in which these are candidate primitives. Fourier transforms, wavelets, autocorrelation, cyclostationary analysis, demodulation and state-space estimation can all be representation proposals. The novelty question is therefore not whether any individual transform is new, but whether a general discovery architecture can select and combine such transforms without a declared carrier model.

3.9 The gap. The literature contains strong solutions to pieces of the problem: universal steganalysis within carriers, atypicality, compression-based discovery, representation learning, multimodal fusion, and blind signal analysis. The gap targeted here is their systematic composition around a carrier-agnostic inverse problem: *discover the representation in which arbitrary observed structure is statistically defensible and potentially recoverable*. This is deliberately framed as a research hypothesis rather than an established novelty claim.

4. Conceptual Model: From Artifact to Canonical Information

Let X be an observed artifact. Let Ω be a family of candidate representation operators. Each R∈Ω maps X into a state sequence S_R=R(X). A decoder D_R maps the state sequence to a candidate canonical object ■_R. The system evaluates a representation using evidence E(R|X), complexity L(R), reproducibility across observations, and—when a reference C exists—recovery quality.

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A useful idealized objective is:
R* = argmax_R [ I(C; ■_R) − λ L(R) − μ D(X, X_R) ]
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where X_R is a reconstruction or synthesized artifact under the candidate representation. If C is unknown, the mutual-information term is replaced by measurable proxies such as predictive compression gain, held-out likelihood, cross-modal consistency, or recoverability of known synthetic probes.

The architecture separates five questions that are often conflated. **Detection:** is there evidence that the observation is non-null under a specified model? **Representation:** which state variables make the evidence stronger? **Channel:** which subset or combination of variables carries the recoverable dependency? **Decoding:** can those states be mapped to a canonical symbol or information sequence? **Residual:** after decoding one channel, does additional structured information remain?

The residual question is essential for multiplexed or hierarchical systems. Suppose an artifact contains C=C1⊕C2⊕C3, with each component projected into a different observable subspace. A detector that finds C1 and terminates has not discovered the information structure of X; it has discovered one explanation. UISA therefore treats search order and stopping criteria as first-class parts of the model.

A candidate channel can also contain routing information. For example, a readily detectable sequence may specify the symbol table, frequency band, offset, or representation needed to decode a second channel. This is not exotic from a systems perspective: protocols routinely use headers to identify how subsequent data should be interpreted. The universal detector must therefore allow discovered structure to modify the subsequent hypothesis space.

A further distinction is required between a *channel* and a *carrier*. A carrier is the observable medium; a channel is a dependency through which information can be represented. The same carrier can contain multiple channels, and the same information can be projected across multiple carriers. This distinction motivates multimodal and cross-domain analysis rather than a fixed media taxonomy.

4A. Mathematical Propositions and Proofs

Proposition 1 (finite-family maximum-score control). Let R_1,...,R_K be a finite family of candidate representations. Suppose each candidate produces a valid null p-value p_k, so that under the null hypothesis P(p_k <= u) <= u for all u in [0,1]. Then the rule min_k p_k <= alpha/K controls the family-wise error rate at level alpha, without requiring independence among candidates.

Proof. Let A_k={p_k <= alpha/K}. By the union bound, P(union_k A_k) <= sum_k P(A_k) <= K(alpha/K)=alpha. Therefore the probability of at least one false discovery among the K searched candidates is at most alpha. Independence is not required. The result is elementary but important for UISA: a significance threshold calibrated for one representation cannot be reused unchanged after searching many representations.

Proposition 2 (complexity-weighted discovery bound). Let each candidate representation R in a countable family Omega be assigned a nonnegative weight pi(R) such that sum_R pi(R) <= 1. Suppose p_R is a valid null p-value for candidate R. Define the discovery score S_R = -log(p_R/pi(R)). Then, for any t>0, P_0(exists R in Omega : S_R >= t) <= exp(-t).

Proof. The event S_R >= t is equivalent to p_R <= pi(R) exp(-t). By p-value validity, P_0(S_R >= t) <= pi(R) exp(-t). Applying the union bound over the countable family gives P_0(exists R:S_R>=t) <= exp(-t) sum_R pi(R) <= exp(-t). The weight therefore acts as an explicit complexity budget. A representation requiring a larger prior/code weight must provide proportionally stronger evidence to win the search.

Corollary 2.1. If pi(R)=2^(-L(R)) for a prefix code of representation descriptions, then S_R = -log(p_R) + L(R)log(2). Thus a short representation receives less complexity penalty than a long one, while a complex representation can still win when its evidence is sufficiently strong. This gives a formal bridge between representation discovery and MDL-style complexity control.

Proposition 3 (recoverability requires retained information). Let C be a canonical payload, X an observed artifact generated from C, R a deterministic or randomized representation operator, and D a decoder producing C-hat=D(R(X)). Then I(C;C-hat) <= I(C;R(X)) <= I(C;X).

Proof. For deterministic R and D, C -> X -> R(X) -> C-hat is a Markov chain. For randomized R, include its independent random seed in the representation mechanism. The data-processing inequality gives I(C;C-hat) <= I(C;R(X)) and I(C;R(X)) <= I(C;X). Hence a decoder cannot recover information discarded by the representation. This is why UISA must evaluate canonical recovery rather than anomaly or compression score alone.

Proposition 4 (search order is a resource-allocation variable). Suppose candidate representations have unequal evaluation costs c(R)>0 and the detector has finite computational budget B. If the search policy evaluates only a prefix whose cumulative cost is at most B, changing the ordering can change the set of evaluated representations and therefore the best representation discovered, even when the candidate family and scoring function are unchanged.

Proof. Choose R_a and R_b such that each is individually affordable but c(R_a)+c(R_b)>B. An ordering beginning with R_a can exhaust the budget before R_b is evaluated; the reverse ordering can do the opposite. If their scores differ, the discovered maximum differs. Therefore search order is part of the detector specification under finite resources. A universality claim must report proposal policy, pruning, ordering, stopping rule and computational budget.

Proposition 5 (residual search can increase recoverable information). Let an artifact contain two conditionally separable channels X=(X_1,X_2) carrying payloads C_1 and C_2. If a first decoder recovers C_1 from X_1 and constructs a residual X_2 that preserves the conditional distribution of C_2 given X_1, then a second-stage search on X_2 can recover information about C_2 that a one-shot decoder restricted to X_1 cannot recover.

Proof. Under the stated separation, I(C_2;X_2|X_1)=I(C_2;X_2) when X_1 provides no additional information about C_2 beyond the residual construction. A decoder restricted to X_1 has no access to this conditional information, whereas a decoder operating on X_2 can attain positive I(C_2;X_2) whenever the second channel is informative. Residual analysis can therefore expose information inaccessible to greedy first-stage search.

Scope. These propositions do not prove universality. They establish mathematical constraints on any candidate universal system: searched hypotheses require search-aware calibration; representation complexity must be accounted for; successful decoding requires information retention; finite budgets make search policy consequential; and residual analysis can be necessary for multiplexed structure. The present prototype implements only a subset of these principles, so the results are design theorems for the research programme rather than empirical claims about the current detector.

5. Statistical Foundations and the Null-Model Problem

The most dangerous failure mode of UISA is confusing “structured” with “encoded.” Natural artifacts are structured. Human language has syntax; music has rhythm and tonal organization; video has physical continuity; engineered machines have control laws. A candidate detector that rewards compressibility will therefore find strong evidence almost everywhere.

A valid significance statement must be conditional on the null model and the search procedure. If a system evaluates K candidate representations and reports the maximum score, the null distribution must be defined for that maximum, not for one preselected representation. Otherwise the system commits a multiple-testing error: searching many hypotheses makes extreme scores likely even when every individual hypothesis is null.

One practical solution is a nested surrogate procedure. Generate null observations that preserve selected nuisance structure while destroying the hypothesized information dependency. For example, a marginal null can preserve the amplitude distribution while randomizing temporal order; a temporal null can preserve local smoothness while breaking long-range structure; a cross-modal null can preserve each modality independently while breaking synchronization. Evidence that survives increasingly strong nulls is more informative than evidence against only an iid null.

The same logic applies to the candidate representation. If the detector searches over frequency bands, offsets, symbol tables, transforms, temporal windows and cross-modal pairings, the null must include the full search. The benchmark therefore treats the maximum candidate score as the statistic of interest and uses surrogate observations to estimate its distribution.

Repeated observations provide another control. A structure that appears in one video may be accidental. A structure that recurs across independently generated or independently observed instances is stronger evidence. The system should therefore accumulate evidence over a series rather than treating one artifact as an absolute decision boundary.

The prototype deliberately does not claim formal p-values. Its circular-shift surrogate is a compact engineering null and is insufficient for all signal classes. The negative result should be read as a methodological warning: even when a score is statistically unusual relative to a surrogate, it does not follow that the underlying structure is an intentional

payload.

A future UISA benchmark should report at least: false-positive rate under multiple nested nulls; detection power as a function of payload strength; performance under representation mismatch; calibration error; search-budget dependence; reproducibility across independent artifacts; and recovery quality after the detector identifies a channel. This moves the evaluation from binary steganalysis toward scientific inference over representations.

6. UISA Architecture

The proposed architecture is organized as a progressive search pipeline rather than a single classifier. The input is an arbitrary artifact or synchronized collection of artifacts. The first stage performs normalization and modality inventory when possible, but does not assume that the declared file type is the effective information carrier.

The second stage constructs a representation bank. Candidate transforms may include identity, temporal difference, Fourier magnitude, spectral-band energy, autocorrelation, wavelets, spatial gradients, optical-flow-like motion states, pose/state trajectories, symbolic segmentation, and cross-modal synchronization. The bank is expandable; it is not intended to be exhaustive.

The third stage computes evidence for each candidate. Evidence can combine compression gain, predictive likelihood, mutual information, transition regularity, cross-modal coherence, recurrence, and learned discriminative features. Each score must be normalized relative to an appropriate surrogate distribution and penalized for representation complexity.

The fourth stage performs channel pursuit. Instead of selecting only the top representation, the system may retain several hypotheses and test whether their residuals contain additional structure. This is analogous to iterative source separation or residual coding: after explaining one component, the unexplained residual becomes the next object of analysis.

The fifth stage attempts canonical recovery. A representation is not considered fully useful merely because it produces a high score. It should map into a compact symbol sequence, known reference, or reproducible semantic object. In synthetic experiments, the injected payload supplies the ground truth; in future natural experiments, recovery may be evaluated using known transformations, human labels, or cross-instance consistency.

The final stage evaluates residual structure and stopping. A practical stopping rule should compare the best remaining evidence against the complexity cost and an independently estimated null. The system should not be allowed to continue decoding indefinitely merely because finite data always permit some model to fit.

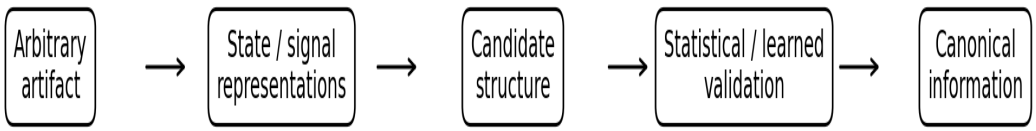


Figure 1. Conceptual UISA pipeline: arbitrary artifact → candidate representations → statistical/learned evidence → channel pursuit → canonical recovery → residual analysis and stopping.

The architecture also supports hardware specialization. Representation discovery and learned feature generation are naturally GPU-friendly because many candidate transforms can be evaluated in parallel. Once a stable representation family has been discovered, low-latency inference primitives can be compiled for FPGA or other accelerators. The distinction mirrors the difference between exploratory analysis and deployed measurement instrumentation.

7. Prototype Implementation

The released prototype is intentionally small. It is designed to make the research question executable rather than to pretend that a complete universal system already exists. The implementation contains six principal components.

Component	Function
core.py	Quantization, entropy, zlib-based description length, and circular-shift atypicality score.
representations.py	Identity, temporal difference, FFT magnitude, spectral-band energy, autocorrelation, and lightweight state proxies.
synthetic.py	Deterministic motion and audio carriers with optional structured-channel injection.
detector.py	Candidate representation scan and surrogate scoring.
features.py / ml.py	Compact spectral-temporal features and cross-modality random-forest baseline.
run_experiment.py	Fixed-seed benchmark generation and machine-readable JSON output.

The candidate detector first quantizes a signal and evaluates a small representation bank. For each candidate, a description-length-inspired statistic is compared with scores obtained from circularly shifted surrogates. The top candidate is retained as the maximum statistic. This is deliberately simple: the point is to expose the statistical behavior of representation search before adding a deep model.

The synthetic generator produces two families: motion-like state trajectories and audio-like signals. Each can be generated in a null form or with a deliberately structured injected channel. The same generator family is used for the learned baseline's train and held-out test distributions, so the resulting performance is a controlled baseline rather than a universal generalization claim.

The learned model is a conventional random forest over compact signal features. It jointly receives motion and audio examples, allowing the benchmark to test whether a classifier can distinguish the structured family when modality is not treated as a fixed separate task. The purpose is not to establish state-of-the-art classification performance; it is to show that the research scaffold can support learned detection and held-out evaluation.

8. Experimental Protocol

The experiment was designed to answer two narrow questions. First, does a simple MDL-inspired representation search distinguish naturally structured carriers from deliberately structured carriers? Second, can a learned feature combination detect the synthetic structured family on independently generated held-out samples? The first question is intentionally adversarial to the proposed method: if the answer were trivially yes, the need for a more principled UISA architecture would be less clear.

For the candidate scan, four conditions are evaluated: motion-null, motion-structured, audio-null, and audio-structured. Twenty seeds are generated for each condition. For each artifact, 24 circular-shift surrogates are used to estimate the score distribution. The benchmark therefore contains 80 primary artifacts and 1,920 surrogate evaluations before accounting for the internal representation bank.

The statistic reported is the maximum candidate atypicality z-score. This choice deliberately captures the multiple-representation search effect. The mean and standard deviation across the 20 seeds are reported for each condition. No threshold is selected post hoc to manufacture a positive result.

The learned baseline uses a fixed random seed and an independently generated held-out test set. Performance is summarized by accuracy and ROC-AUC. Because the synthetic generator family is shared, the experiment does not test open-world generalization. It tests whether the feature stack contains enough information to detect the known structured family under a clean train/test split.

The benchmark was re-run during preparation of this extended draft. The unit test suite passes 2/2. The benchmark reproduces the same four candidate-scan means and standard deviations and the same random-forest accuracy/AUC as the earlier draft. The machine-readable benchmark JSON was regenerated rather than copied forward.

Validation item	Result
Unit tests	2 passed
Motion null	2.044 ± 0.386
Motion structured	2.058 ± 0.475
Audio null	0.797 ± 0.639
Audio structured	1.135 ± 0.878
RF accuracy	0.825
RF ROC-AUC	0.916328125
Benchmark SHA-256	462d3f88ef60e035d220e784bb1bc3d1cfb8c4b6e380ea8c6e4cedfb627a1e43

The exact benchmark artifact is included in the research package. The hash above is a fixity identifier for the regenerated JSON at the time of this draft. It is useful when comparing a later Zenodo version against this research record.

9. Results

9.1 Candidate-search result. The central empirical result is the overlap between null and structured conditions. Motion-null data have a mean maximum z-score of 2.044±0.386, while motion-structured data have 2.058±0.475. The difference is negligible relative to the dispersion. Audio shows a somewhat larger separation, with 0.797±0.639 for null data and 1.135±0.878 for structured data, but the distributions remain strongly overlapping.

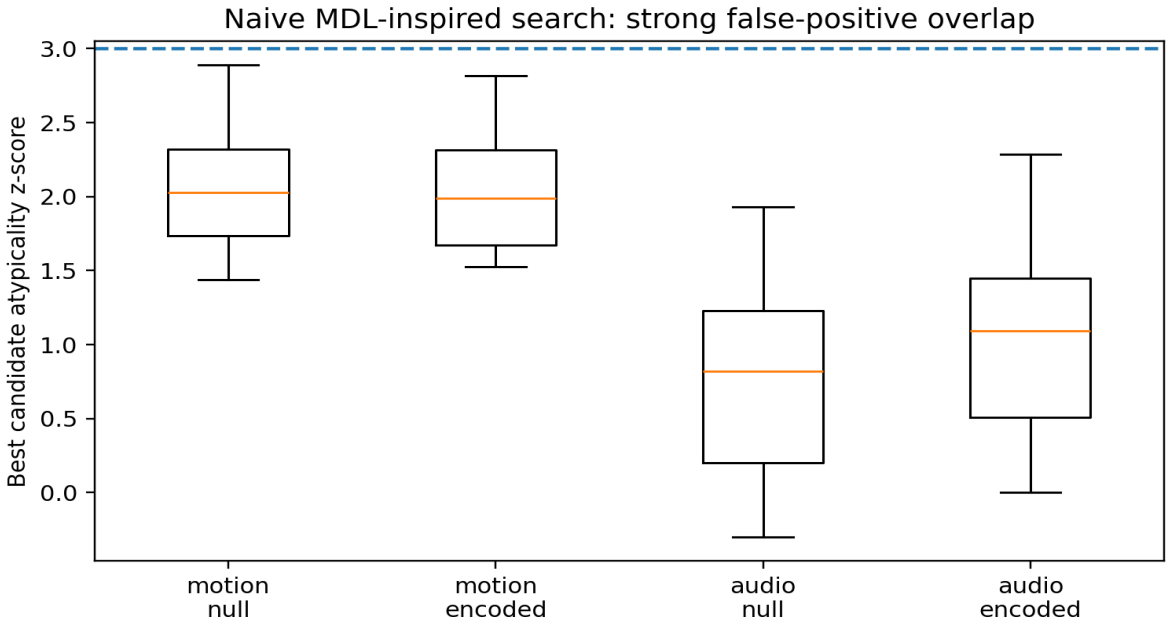


Figure 2. Maximum candidate atypicality distributions. Natural and deliberately structured conditions overlap substantially. This is a negative result for the hypothesis that generic description-length-based anomaly detection is sufficient.

This result is scientifically useful. A natural carrier is not an information-free null. Smooth motion, periodic audio and other ordinary processes possess substantial regularity. A compressibility or atypicality detector will therefore often find them interesting. The experiment demonstrates why a universal information detector must distinguish at least three concepts: ordinary structure, unusual structure, and recoverable encoding.

9.2 Learned baseline. The cross-modality random forest reaches 0.825 accuracy and 0.9163 ROC-AUC. The held-out ROC curve is shown below.

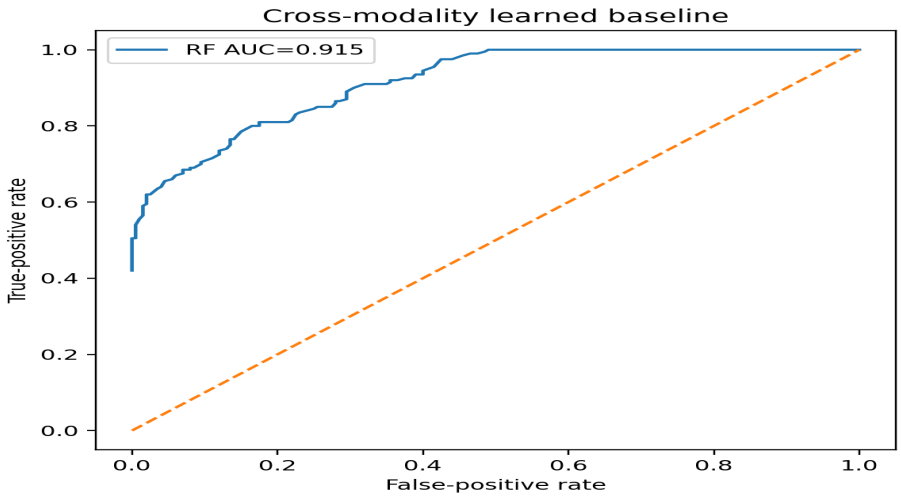


Figure 3. Held-out ROC curve for the deliberately narrow learned baseline.

The learned result demonstrates feasibility, not universality. Because training and test examples share the same synthetic generator family, the classifier can learn generator-specific cues. A genuinely carrier-agnostic benchmark must test on unseen signal families, unseen transformations, and preferably real-world data.

9.3 Strength ablation. The strength experiment examines how the candidate score changes as the injected channel becomes stronger. The response is gradual rather than a clean step function. This is consistent with the observation that injected structure is superimposed on pre-existing carrier structure.

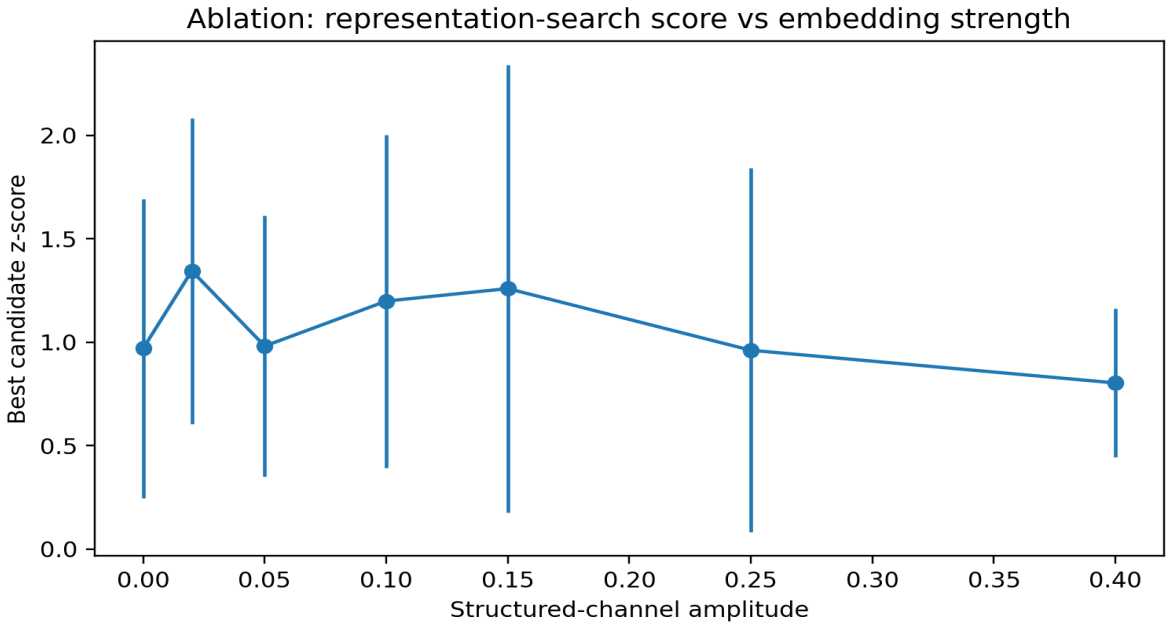


Figure 4. Embedding-strength ablation for the audio carrier. The result illustrates gradual interaction between carrier structure and injected structure.

10. What the Negative Result Means

It is tempting to interpret the candidate-scan overlap as evidence that the UISA idea is impractical. That interpretation would be incorrect. The experiment was not a test of a complete UISA detector; it was a test of a particular low-complexity score. The score asks, approximately, whether a candidate representation is more atypical/compressible than its circular-shift surrogates. It does not ask whether the discovered regularity maps to a stable symbol alphabet or whether a message can be reconstructed.

This distinction is central. Suppose a dancer repeats a culturally conventional movement sequence. A representation based on joint angles may show strong temporal structure. A representation based on beat phase may show even

stronger structure. Both are real information-bearing descriptions, but neither implies that an external message was encoded. Conversely, an intentional payload could be embedded in a relation that is statistically weak in every single marginal channel but strong in a cross-channel transition. A detector that stops at marginal structure will miss it.

The appropriate response is therefore not to seek one magic statistic. The detector must maintain a hypothesis graph. Each node represents a candidate representation or channel; edges represent transformations or dependencies between nodes. Evidence can be accumulated along paths. A high-scoring node may become a routing hypothesis for another node. Residual analysis can remove the explained component and reopen the search.

This also changes how “false positive” should be defined. A natural structure is not necessarily a false positive for the claim “this artifact contains structure.” It is a false positive only for the stronger claim “this artifact contains an intentional recoverable payload.” These hypotheses require different labels and different experiments.

A mature benchmark should therefore contain at least four labels: *null* (no injected or target structure), *natural structure* (ordinary organized artifact), *synthetic recoverable structure* (known encoding), and *ambiguous/unknown*. The fourth category matters for real data because a detector should be allowed to report “structured but not decoded” rather than forcing a binary decision.

The negative result thus establishes a design constraint: UISA must be a layered inference system. A one-stage classifier is insufficient as the scientific definition of the problem, even if it can perform well on a narrow benchmark.

11. Representation Discovery as the Core Research Problem

The strongest formulation of UISA is not “find hidden messages.” It is “discover useful coordinates.” Let X be an artifact and let Ω be a family of transformations. The system seeks $R \in \Omega$ such that R(X) has properties that are difficult to explain under the null but useful for recovery. Those properties may include compression, prediction, synchronization, recurrence, separability, or cross-modal consistency.

This makes UISA closely related to scientific model discovery. A representation is valuable when it reduces the description of observations without merely memorizing them. The complexity penalty is therefore not optional. A representation that includes a lookup table containing the entire input is a perfect decoder and a useless scientific explanation.

The practical search space can be constructed compositionally. Primitive transforms form the first layer. Their outputs can be quantized into symbols. Symbol sequences can be segmented, clustered, or mapped into candidate alphabets. Candidate alphabets can then be tested against transition models. Cross-modal operators can combine two or more representations. Finally, discovered structures can recursively define the next search stage.

This compositionality is important for scalability. A universal system cannot enumerate arbitrary programs, but it can search a library of reusable primitives and learn how to compose them. GPU acceleration is particularly attractive here: hundreds or thousands of candidate transforms can be evaluated in parallel, while a learned controller ranks promising hypotheses.

The system should also expose uncertainty. Instead of producing one decoded answer, it can maintain a posterior or ranked hypothesis set: representation, channel, synchronization, alphabet, confidence, residual score and complexity. Human review can then inspect the most informative hypotheses. This is closer to an instrument that supports discovery than a black-box classifier that outputs a single label.

An especially important future direction is learned representation proposal. Rather than training a network to recognize one carrier, train a model to propose transformations that maximize held-out compression or predictive recovery under a complexity budget. The network becomes a search policy over representations rather than a fixed detector.

12. Multiplexing, Hierarchical Channels, and Detector Search Order

A general information-bearing artifact may contain several simultaneous channels. They need not be independent. One channel may carry a payload, another may carry synchronization, and a third may carry the index of the representation required to decode the first. The channels can also overlap in physical coordinates while remaining separable in a learned state space.

This motivates a formal distinction between **channel existence** and **channel discovery order**. A detector has a finite computational budget. If it finds a highly salient channel early, a greedy system may stop. But salience and importance are not equivalent. A low-amplitude cross-modal relation may contain more information than a dominant periodic

carrier feature.

The search policy should therefore maintain multiple hypotheses. One possible strategy is beam search over representation graphs, where each candidate has a score composed of evidence gain minus complexity and compute cost. Another is sequential experimental design: select the next transformation expected to maximally reduce uncertainty about the remaining hypotheses.

The concept also clarifies the role of decoy structure. A deliberately salient channel can cause a detector to terminate prematurely even though additional channels remain. From the standpoint of UISA, this is not a special property of steganography; it is a generic adversarial property of search procedures. The correct response is to make residual analysis and stopping criteria explicit.

A rigorous benchmark should therefore include multiplexed synthetic artifacts where payloads are distributed across time, frequency, spatial position, modalities, or combinations thereof. The evaluation should report not only whether any channel is detected but how many channels are recovered, in what order, with what payload fidelity, and how much computation is consumed.

The same framework naturally handles physical state trajectories. A dancer can be represented as S_t=[q_t, v_t, a_t, g_t, p_t, b_t, u_t,…], where q denotes joint configuration, v velocity, a acceleration, g gaze, p stage position, b beat phase, and u additional state variables. The information-bearing object is then not “the video” but the trajectory and its projections. This provides a concrete bridge from digital media to arbitrary engineered and physical systems.

12A. Information Channels, Arbitrary Alphabets, and Representation Families

12A.1 Arbitrary symbol systems. A central motivation for UISA is the observation that an information alphabet need not consist of primitive characters. In a conventional digital system, the alphabet may be bytes, Unicode code points, MIDI events, pixels, or sensor quantization levels. But an engineered encoding can instead assign symbols to composite objects: phrases, musical motifs, visual configurations, motion primitives, biological sequences, or complete state transitions. A symbol can therefore have internal structure that is itself meaningful in another representation. The inverse problem is consequently recursive: the analyst may need to discover not only which states constitute symbols, but which substructure should be treated as atomic for the decoding task. This motivates a representation lattice rather than a flat list of transforms. A candidate can be refined into subrepresentations or composed with another candidate. Longest-match parsing, hierarchical tokenization, and canonicalization become relevant once composite symbols are admitted.

12A.2 Projection rather than carrier identification. The distinction between projection and carrier is useful because the same information can be mapped into many observable domains. A textual object may be projected into musical durations; a sequence may be projected into a trajectory; a trajectory may be projected into video; and the resulting artifact can retain the original information while appearing to belong naturally to the target medium. In this setting, asking whether the carrier is “really” audio or video is less informative than asking which projection preserves a recoverable dependency. UISA therefore treats modality labels as metadata supplied by an observer rather than as ontological restrictions on the information process.

12A.3 Physical systems as information spaces. The same reasoning extends beyond conventional media. A mechanical mechanism has positions, velocities, contacts, forces, timing relationships and constrained transitions. A biological system has sequence states, expression states, morphology and temporal trajectories. A dancer has joint configurations, hand configurations, gaze, orientation, stage position, timing and relationships to music. These systems can all be sampled into state trajectories. The important research question is then whether a transformation of the trajectory exposes a lower-complexity, predictive, or recoverable symbolic process. This is not a claim that every physical process is an encoded message; it is a proposal that physical state should not be excluded from information analysis merely because its native representation is not a conventional digital alphabet.

12A.4 Multiplexing and residual search. Once arbitrary channels are admitted, a greedy detector is insufficient. Let an observation contain $C = C_1 \oplus C_2 \oplus C_3$, where each component occupies a different projection or dependency. Detecting C_1 and declaring success loses information about the residual. A mature UISA system should decode a candidate channel, reconstruct or subtract its explanatory contribution, and repeat the search on the residual. The process resembles iterative source separation, but with representation hypotheses and canonical recovery included in the loop. The stopping condition should depend on calibrated evidence and complexity rather than on the discovery of the first salient pattern.

12A.5 Routing channels and self-describing representations. A discovered channel may also contain instructions about how another channel should be interpreted. A detectable component could specify an offset, symbol table, frequency interval, synchronization rule, or representation transform. This creates a hierarchy in which one recovered representation expands or contracts the subsequent hypothesis space. Such behavior is already familiar in communication protocols, where headers and metadata determine how payload bytes are parsed. The difference here is that the protocol is not given to the receiver in advance. UISA must therefore allow evidence from one representation to become a proposal mechanism for another while retaining an explicit complexity penalty and held-out validation.

12A.6 Search as an experimental object. The detector's search order is part of the scientific object. Two systems given the same artifact and the same candidate library can produce different discoveries if one searches frequency-domain representations first and another searches temporal transitions first. Consequently, claims of universality must specify the proposal distribution, pruning rules, early-stopping policy, residual construction, and computational budget. A universal system cannot be evaluated only by asking whether a correct representation exists somewhere in an enormous hypothesis space; it must show that its search procedure can find useful representations within a defined resource budget.

12A.7 Relation to the constructive encoding problem. UISA also forms the inverse counterpart to a constructive encoding framework. In a forward system, canonical information is projected into an arbitrary symbol space or physical carrier. In the inverse system, the observation is given and the representation must be discovered. This symmetry is scientifically useful because a controlled forward encoder can generate ground-truth data for the inverse detector. The resulting benchmark can vary symbol alphabets, synchronization, channel location, modality, redundancy, transformations and payload strength while retaining exact knowledge of the canonical object. This creates a principled path from toy demonstrations to measurable representation-discovery experiments.

12B. Human–AI Research Method and Scientific Accountability

12B.1 Collaboration as a research process. The present work was developed through iterative human–AI reasoning in which the human researcher supplied the problem direction, conceptual constraints, counterexamples, domain knowledge and decisions about research significance, while the AI system assisted with formalization, literature synthesis, implementation, experimental scaffolding, criticism and manuscript construction. This mode of collaboration creates a special reproducibility requirement: the final artifact must distinguish the provenance of ideas from the provenance of executable evidence. A polished narrative is not evidence that an experiment was run, and an AI-generated implementation is not evidence that the implementation is correct. The research record therefore treats code execution, machine-readable results, tests and hashes as the authoritative layer for numerical claims.

12B.2 Intellectual provenance. A human–AI paper should preserve the trajectory by which the research question became precise. In the present case, the problem evolved from constructive arbitrary information projection toward its inverse: discovering recoverable structure when the representation, carrier and alphabet are not known. Intermediate discussions about music, video, physical motion, multiplexing, offsets, symbol tables and statistical outliers are not merely illustrative examples; together they motivated the separation of carrier, channel, representation and canonical information. A complete research record should therefore retain these conceptual bridges while distinguishing them from experimentally established results.

12B.3 Literature synthesis rather than citation accumulation. The literature review is organized by assumptions and problem boundaries. Shannon theory supplies the source/channel vocabulary; MDL supplies complexity-aware model comparison; atypicality supplies a coding-based notion of interestingness; steganalysis supplies detection of hidden structure within declared carriers; blind signal processing supplies transforms and source-separation primitives; representation learning supplies learned coordinate systems; and multimodal learning supplies mechanisms for joint evidence. The proposed framework is not justified merely by listing these fields. Its research question arises from relaxing their usual assumption that the relevant representation or carrier family is known before analysis.

12B.4 Evidence hierarchy. The manuscript adopts an explicit hierarchy of claims. A statistical pattern is weaker than a reproducible pattern; a reproducible pattern is weaker than a recoverable symbol stream; a recoverable stream is weaker than a validated canonical payload; and none of these alone establishes human intent. This hierarchy prevents the common failure in which an anomaly detector is interpreted as a decoder. It also makes negative results informative: if a representation is statistically salient but cannot recover a known payload, it is evidence of structure but not evidence of successful information extraction.

12B.5 Manual review requirements. The paper is deliberately constructed so that a researcher can audit the central claims without trusting the generated prose. Equations state the optimization target; the benchmark records retain individual observations; the source tree contains the candidate transformations; the tests verify basic computational invariants; the validation report records reproduction status; and the limitations state where the experiment is insufficient. For a subsequent peer-reviewed submission, every empirical statement should remain traceable to either a source citation or a reproducible artifact. This is particularly important when AI assistance is used because fluent text can conceal unsupported transitions between assumptions and conclusions.

12B.6 Zenodo as the first public research record. The first release is intended to establish a stable public research object before venue-specific submission. The release should identify the exact manuscript version, code, data, figures and validation state. Later improvements should be versioned rather than silently replacing the artifact. The Zenodo record should make clear that public deposition establishes precedence and reproducibility but does not imply peer-review acceptance. A later conference or journal manuscript may refine the exposition, expand the experiments, anonymize the submission, or change the formatting while citing the versioned research record as the earlier public artifact.

12B.7 General rule for human–AI scientific writing. The governing principle is that AI assistance should increase the inspectability of the research rather than decrease it. The system may help generate prose, equations, code, figures and experimental plans, but the final artifact must preserve human responsibility for research questions, interpretation, citations, experimental validity and release decisions. In particular, no numerical result should enter the manuscript merely because it appears plausible in generated prose. Every such result must be regenerated or traced to a recorded artifact. This rule is applied in the present revision: the benchmark was re-executed, the unit tests were rerun, and the manuscript was rebuilt from the resulting research tree.

13. Computational Complexity and Engineering Considerations

A common mistake in discussing universal decoding is to equate the number of hypotheses with the asymptotic runtime of the decoder. These are distinct. For an N-byte input there are 2^(8N) possible byte strings, but a practical decoder does not enumerate all byte strings. It searches a restricted hypothesis space. If an unknown offset has N possible positions and each candidate scan costs O(N), the total can be O(N^2), even though the offset hypothesis count is only O(N). Conversely, an O(1) algorithm can have a very large constant factor and be slower than an O(N) algorithm for practical N.

UISA therefore needs at least three complexity measures: hypothesis-space size H, evaluation cost C(R,X) for a candidate representation, and total search cost Σ_R C(R,X). Learned proposal mechanisms can reduce the effective number of candidates, but they do not make the underlying problem free. They shift computation from explicit enumeration into model inference.

GPU execution is appropriate for broad candidate evaluation. FFTs, convolutions, feature extraction, surrogate generation and neural representation proposals are highly parallel. The GPU can operate as a discovery engine that scores many hypotheses simultaneously. Once a representation family stabilizes, an FPGA or specialized accelerator can implement the deterministic subset at low latency and power.

A deployment architecture could therefore have two layers. The exploratory layer is adaptive, probabilistic and compute-intensive. The compiled layer contains validated representations, synchronization rules and decoders. A newly discovered representation can move from exploratory status to compiled status only after reproducibility tests. This creates an engineering analogue of scientific instrument calibration.

Memory bandwidth may become a stronger bottleneck than arithmetic throughput for multimodal artifacts. The architecture should avoid repeatedly materializing all transformed representations. Streaming transforms, feature caches, approximate search and hierarchical pruning become important for long video and high-rate sensor data.

The prototype does not yet benchmark these engineering dimensions. They are included here because a universal analysis system that cannot operate on realistic data rates would remain a conceptual framework. The next implementation cycle should therefore report throughput in samples/s or frames/s, latency per candidate, memory footprint, GPU utilization, and energy per analyzed unit.

14. Limitations and Threats to Validity

The first limitation is scope. The current system does not analyze arbitrary real-world artifacts. It contains a small hand-designed representation bank and synthetic carriers. The term “universal” is consequently a research objective,

not an empirically established property.

The second limitation is the null model. Circular shifts are appropriate for some stationary or approximately stationary signals but can be invalid for strongly nonstationary processes. A dancer, for example, has meaningful temporal progression, and circular shifting can introduce artifacts that make the null too easy or too hard. Future experiments require block permutations, phase randomization, matched surrogates and generative null models.

The third limitation is generator leakage in the learned baseline. Training and testing use the same broad synthetic generator family. The AUC of 0.9163 therefore cannot be interpreted as carrier-agnostic detection. It only establishes that the selected feature family contains predictive information for the benchmark distribution.

The fourth limitation is the absence of canonical recovery. Detection is easier to define than decoding. A future system must demonstrate that a discovered representation produces a stable symbol stream or information object, and that the recovered object survives independent validation.

The fifth limitation is multiple testing. The current prototype reports maximum candidate scores but does not provide a fully corrected inferential procedure for an arbitrarily expanding representation library. A mature implementation needs hierarchical testing, held-out search budgets, or a fully specified MDL/Bayesian model-selection framework.

The sixth limitation is semantic ambiguity. A statistically significant pattern may be meaningful to a human, meaningful only to the encoder, or accidental. UISA can provide evidence and candidate decodings; it cannot infer human intent from statistics alone. Intentionality requires external evidence, controlled generation, or an agreed semantic criterion.

Finally, the current work does not establish superiority over state-of-the-art steganalysis within any particular carrier. Such comparisons would be appropriate only after the universal problem is operationalized carefully enough to permit fair modality-specific baselines.

15. Research Roadmap and Proposed Experiments

The next stage should expand the benchmark along four axes: carrier diversity, representation diversity, channel diversity, and transformation robustness.

Stage	Experiment	Primary metric
A	Natural text, music, speech, video, dance and mechanical trajectories with matched surrogates	False-positive calibration across nulls
B	Unknown frequency/time/spatial/symbolic channels	Detection power vs. payload strength
C	Multiplexed and hierarchical channels	Number and fidelity of recovered channels
D	Cross-modal split payloads	Incremental information from joint analysis
E	Lossy transformations: resampling, filtering, codecs	Recovery after transformation
F	Unseen generator families	Open-world generalization
G	GPU implementation	Throughput, latency, memory
H	FPGA/compiled decoder	Latency and energy per analyzed unit

A particularly strong benchmark would use paired datasets. Each natural artifact would be transformed by several independently designed encoders, while matched controls would preserve the same surface statistics but remove the recoverable dependency. This would allow the system to separate “natural structure” from “known recoverable structure” without relying on crude iid nulls.

Another key experiment is cross-family generalization. Train the representation proposer on one family of signals and evaluate it on entirely different generators. For example, train on audio and motion but test on mechanical sensor streams and synthetic biological trajectories. Success would provide stronger evidence for representation-level generality than another increase in within-family AUC.

Canonical recovery should become the primary endpoint. Given a known payload C, the system should output ■ and report exact recovery, edit distance, mutual information, or task-specific distortion. The detector should be penalized if it finds an attractive representation that cannot recover the payload.

Finally, the system should be evaluated after ordinary transformations. If information is represented in music, resampling and band-limiting should be tested. If represented in video, recompression, frame-rate conversion and scaling should be tested. If represented in physical trajectories, measurement noise and sensor sampling should be tested. This connects the abstract representation problem to actual communication and sensing systems.

16. Reproducibility, Research Artifact, and Zenodo Release

The research package is designed to be deposited as a single reproducible artifact containing the paper, source code, tests, figures, machine-readable benchmark output, and documentation. This is preferable to depositing only a PDF because the scientific object includes the computational procedure that generated the numerical claims.

The benchmark JSON is regenerated during this revision rather than merely asserted to match the previous result. Unit tests pass 2/2. The numerical results reproduce the earlier draft exactly to the reported precision. The SHA-256 digest of the current benchmark JSON is recorded in Section 8 so that a later deposit can distinguish the exact artifact used for this version.

Zenodo supports versioned research records and assigns a DOI to published records. Its documentation distinguishes a version-specific DOI, which should normally be used when citing an exact research artifact, from a concept DOI representing the evolving record. This is useful for the present project because the Zenodo deposit can establish a citable first research record and later versions can preserve the lineage of the code and paper.

For the first public release, the recommended contents are: (1) the extended paper PDF; (2) the complete source tree; (3) the machine-readable benchmark results; (4) figures; (5) a README with environment and reproduction commands; (6) a validation report containing test status and hashes; and (7) an explicit statement that the work is a research prototype and that universality is not yet empirically demonstrated.

The metadata should use a stable title, creator identity, abstract, keywords, software/data descriptors, and a clear license. If later files change, a new Zenodo version should be created rather than modifying the already published file set. This preserves reproducibility for anyone citing the first release.

The present document is therefore best understood as a **research record**, not as a claim of peer-review acceptance. Publication on Zenodo establishes public precedence and a persistent research object; peer review remains a separate process.

Artifact contents at the time of this draft include: UISA_conference_draft.pdf; uisa/ source package; run_experiment.py; make_figures.py; tests; results/benchmark.json; figures; README.md; requirements.txt; pyproject.toml; and RESULTS.md.

17. Conclusion

Universal Information Structure Analysis proposes a broader formulation of a problem that is currently distributed across several research communities. Steganalysis asks whether a declared carrier has been modified by a hidden-communication process. Anomaly detection asks whether an observation is unusual under a model. Signal processing extracts structure from known classes of signals. Representation learning discovers latent coordinates. MDL and atypicality quantify explanatory compactness. UISA asks how these ideas can be assembled into an inverse system that begins with an arbitrary artifact and searches for the representation in which recoverable information becomes statistically and computationally defensible.

The prototype demonstrates why this is not a trivial extension of anomaly detection. The naive candidate scan does not separate natural from deliberately structured signals: motion null and structured conditions have almost identical mean maximum z-scores, and audio conditions overlap substantially. The learned baseline performs substantially better on the synthetic family, but its 0.9163 ROC-AUC is explicitly not a universal result because the train and test distributions share a generator family.

The negative result is therefore central to the paper. It shows that natural structure must be modeled rather than treated as noise, that representation search creates a multiple-testing problem, and that recoverability must be separated from statistical interestingness. A universal detector should not stop when it finds the first salient pattern. It should pursue residuals, cross-modal dependencies, alternative representations and possible routing structures until additional evidence no longer justifies its complexity.

The proposed research program is consequently layered: **detect** → **represent** → **localize channel** → **decode** → **validate** → **subtract** → **search again**. This architecture is compatible with classical signal-processing transforms, learned representation proposals, GPU-scale parallel search, and later FPGA compilation of validated primitives.

At this stage, the strongest scientific claim is not that UISA is already universal. It is that the carrier-agnostic representation-discovery problem can be stated precisely enough to benchmark, implement and falsify. The released artifact provides that starting point. The next decisive step is a real multimodal benchmark in which the system must

discover recoverable structure across previously unseen carriers and transformations.

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Web-verified literature note. The 2026 HEVC universal-video paper is published in Frontiers in Artificial Intelligence (15 September 2026); the audiovisual split-payload work is an arXiv preprint dated 7 June 2026; the 2019 atypicality paper and the 2022 image universal-steganalysis paper were checked against publisher/index records during this revision.

Appendix A. Exact Reproduction Procedure

1. Create a Python 3.10+ virtual environment. 2. Install the packages listed in requirements.txt. 3. Run `pytest -q`. 4. Run `python run_experiment.py`. 5. Run `python make_figures.py`. 6. Inspect results/benchmark.json and compare its SHA-256 digest with the digest recorded in the paper version being reproduced.

The experiment uses fixed seeds for the 20 candidate-scan repetitions and the learned baseline. The result file contains every candidate-scan observation, not merely the aggregate means. This allows later researchers to recompute summaries or apply alternative statistical tests without regenerating the data.

The package intentionally does not hide negative results. The candidate detector’s overlap is preserved in both the machine-readable output and the figures. This is important because the principal scientific lesson of the prototype is that a generic compressibility/atypicality heuristic is not sufficient.

The learned baseline should be reproduced with caution. The synthetic generator family is shared between training and held-out testing. A researcher extending the benchmark should add a generator-disjoint test before interpreting the classifier as a universal detector.

Appendix B. Data Schema

Field	Meaning
kind	motion or audio
structured	whether the synthetic structured channel was injected
seed	generation seed

Field	Meaning
best	highest-scoring candidate representation
all	candidate representation results and surrogate-derived scores

Each benchmark record is therefore traceable to a carrier family, condition and seed. The aggregate result is not a hand-entered table.

Appendix C. Interpretation Guide

Do not interpret the current detector as a universal steganalysis system. It is a prototype representation-search experiment. **Do not interpret 0.9163 AUC as open-world performance.** The synthetic train/test families overlap. **Do interpret the candidate-scan overlap as evidence that natural structure must be modeled. Do interpret the code and benchmark as a reproducible scaffold.**

Appendix D. Research Questions for the Next Version

RQ1: Can a representation proposer generalize to a carrier family absent from training? RQ2: Can the system recover a known canonical payload without knowing the carrier or symbol alphabet? RQ3: Can it distinguish natural structure from recoverable synthetic structure under matched nulls? RQ4: Can it recover multiple channels without greedy early stopping? RQ5: Can the discovered representation survive realistic transformations? RQ6: What is the compute/accuracy trade-off for GPU exploration and FPGA compilation? RQ7: Can statistical evidence be calibrated across a growing representation library?

Appendix E. Revision Statement

This extended version supersedes the earlier six-page conference-style draft in the working package. The benchmark data and code are retained; the paper has been expanded rather than replacing the experimental claims with new data. The numerical claims in the main text are those produced by the regenerated benchmark during this revision.

End of preliminary appendices.

Appendix F. Human–AI Research Provenance

Purpose. This appendix records the provenance model used for the living research document. It distinguishes research direction, conceptual propositions, formalization, implementation, empirical execution and editorial synthesis. The human researcher remains responsible for research questions, interpretation, release decisions and scientific claims; the AI system is recorded as an assisting system that contributed formalization, code, literature synthesis, experimental scaffolding and manuscript construction.

Human-originated directions recorded in the conversation. The research direction included arbitrary information projected into arbitrary symbol/state systems; symbols that can themselves be phrases, poems, stories, musical structures, genes or physical states; music, video and physical performances as information-bearing state trajectories; observable variables such as BPM acting as indices; multiple and hierarchical channels; routing information carried by one channel for another; residual search after an initially detected channel; detector search order as part of the system; higher-order, temporal and cross-modal information; moving beyond the term steganography; discovering representation rather than assuming it; and separating natural structure from recoverable encoded structure.

AI contributions recorded in the research tree. The AI formalized these directions into UISA; synthesized adjacent literature; formulated candidate objectives and null-model requirements; implemented the prototype representation bank, atypicality scoring, synthetic generators, feature extraction and random-forest baseline; generated figures and benchmark reports; drafted and revised the manuscript; and identified limitations and experimental extensions. These contributions are verifiable from the source tree and benchmark rather than inferred from prose alone.

Iterative co-development. Some components emerged through repeated proposal, critique and correction and are not usefully assigned wholly to either party. The formal distinction between carrier, channel, representation and canonical information is one example, as is the distinction between hypothesis-space size, per-hypothesis evaluation cost and total search cost. The provenance principle is therefore *human-directed, AI-formalized, iteratively corrected*.

Prompt provenance. The conversation contains many exploratory prompts and corrections. This appendix records substantive prompt themes rather than fabricating a verbatim transcript that is not embedded in the research artifact. Where exact wording is required for audit, the authoritative source should be the exported conversation record associated with this Zenodo version. This is deliberate: provenance must be accurate, not reconstructed from memory

and presented as quotation.

Evidence hierarchy. Conceptual provenance is supported by the conversation record; implementation provenance by source files; numerical provenance by benchmark JSON and execution logs; literature provenance by cited publications and dated bibliographic records; release provenance by the Zenodo version and repository commit once assigned.

Recommended citation language. “This research was developed through iterative human–AI collaboration. The human researcher directed the research questions, conceptual scope and interpretation; the AI system assisted with formalization, implementation, literature synthesis, experimentation and manuscript construction. Empirical claims are supported by the released computational artifacts.”

Appendix G. Living Research Record and Code Provenance

The paper is a living research document. Its scientific identity is the versioned combination of manuscript, executable code, benchmark data, figures, validation reports and provenance records. The canonical code repository is intentionally left to be assigned in a subsequent revision. Once selected, this appendix should record the repository URL, commit hash, release tag and license.

A future release should preserve the mapping: paper version -> code commit -> benchmark hash -> figure set -> validation report. If code changes after publication, a new version should be created rather than silently changing the code associated with an existing research claim.

Appendix H. Responsible Research and Dual-Use Considerations

Carrier-agnostic representation discovery is a general capability and therefore can have both beneficial and harmful downstream applications. The present work addresses this at the level of research governance rather than by publishing a catalogue of misuse scenarios. The prototype is restricted to synthetic benchmarks and does not establish operational capability against arbitrary real-world communications or physical systems.

The paper intentionally describes the scientific capability, limitations, validation requirements and authorization boundary without enumerating application-specific misuse recipes. The governing principle is **capability transparency without misuse optimization**: describe what is being studied, why it matters, how it is validated, and what it does not establish, while avoiding gratuitous operationalization of harmful applications.

This treatment recognizes an information race: withholding every possible implication does not guarantee that motivated actors cannot derive them, particularly when machine reasoning systems can analyze public literature. Conversely, publishing an operational catalogue of harmful uses can unnecessarily lower the cost of misuse. The appropriate position is transparent scientific description combined with restrained operational detail.

Appendix I. Mathematical Interpretation of the Prototype

The prototype is an estimator over a restricted representation family, not an implementation of the full UISA theory. Its current score is an MDL-inspired heuristic based on computable description lengths and surrogate comparisons. Propositions 1–5 specify requirements that a mature implementation should satisfy. The maximum-over-candidates problem should eventually be calibrated using a max-statistic, a complexity-weighted bound, held-out discovery, or an equivalent formally justified procedure.

The current benchmark does not claim to test these propositions directly. Mathematical validity of a selection rule is not empirical validation of a detector, and empirical performance of the present heuristic is not a proof of universality.

Appendix J. Revision and Release Ledger

Working draft history: (1) initial short conference-style prototype; (2) expanded research record; (3) compacted typesetting revision after manual review; (4) mathematical-strengthening revision adding explicit propositions and proofs plus provenance, living-document and responsible-research appendices. The numerical benchmark is retained rather than replaced merely to improve presentation.

The present revision strengthens the mathematical specification and makes provenance explicit without upgrading the empirical evidence beyond what the existing benchmark supports.

End of living research record.